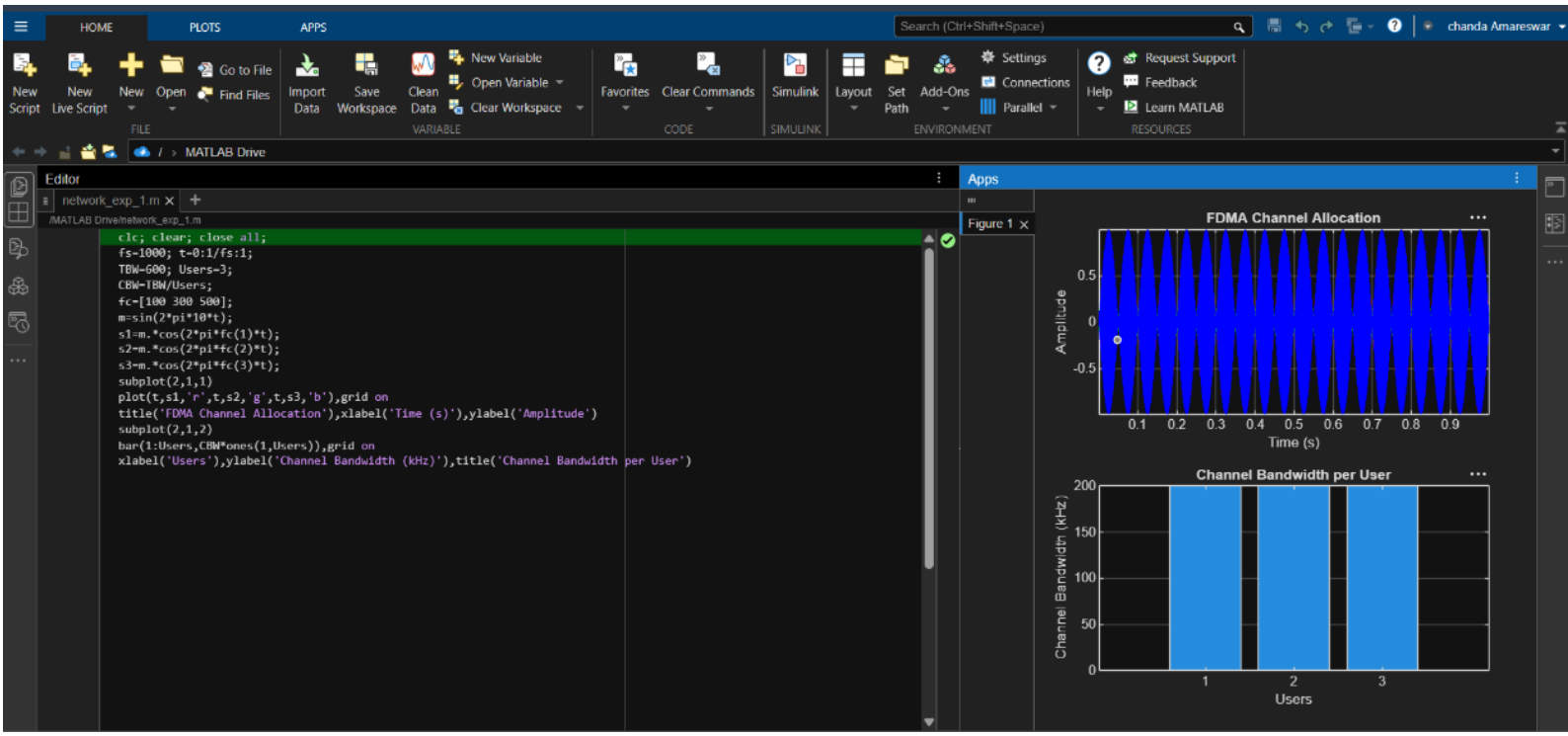
EXPERIMENT 01(OBJ-1):-

PROGRAM CODE:-

```
clc; clear; close all;
fs=1000; t=0:1/fs:1;
TBW=600; Users=3;
CBW=TBW/Users;
 fc=[100 300 500];
 m=sin(2*pi*10*t);
s1=m.*cos(2*pi*fc(1)*t);
s2=m.*cos(2*pi*fc(2)*t);
s3=m.*cos(2*pi*fc(3)*t);
subplot(2,1,1) plot(t,s1,'r',t,s2,'g',t,s3,'b'),
grid on
title('FDMA Channel Allocation'),
xlabel('Time (s)'),
ylabel('Amplitude') subplot(2,1,2)
bar(1:Users,CBW*ones(1,Users)),
grid on xlabel('Users'),
```

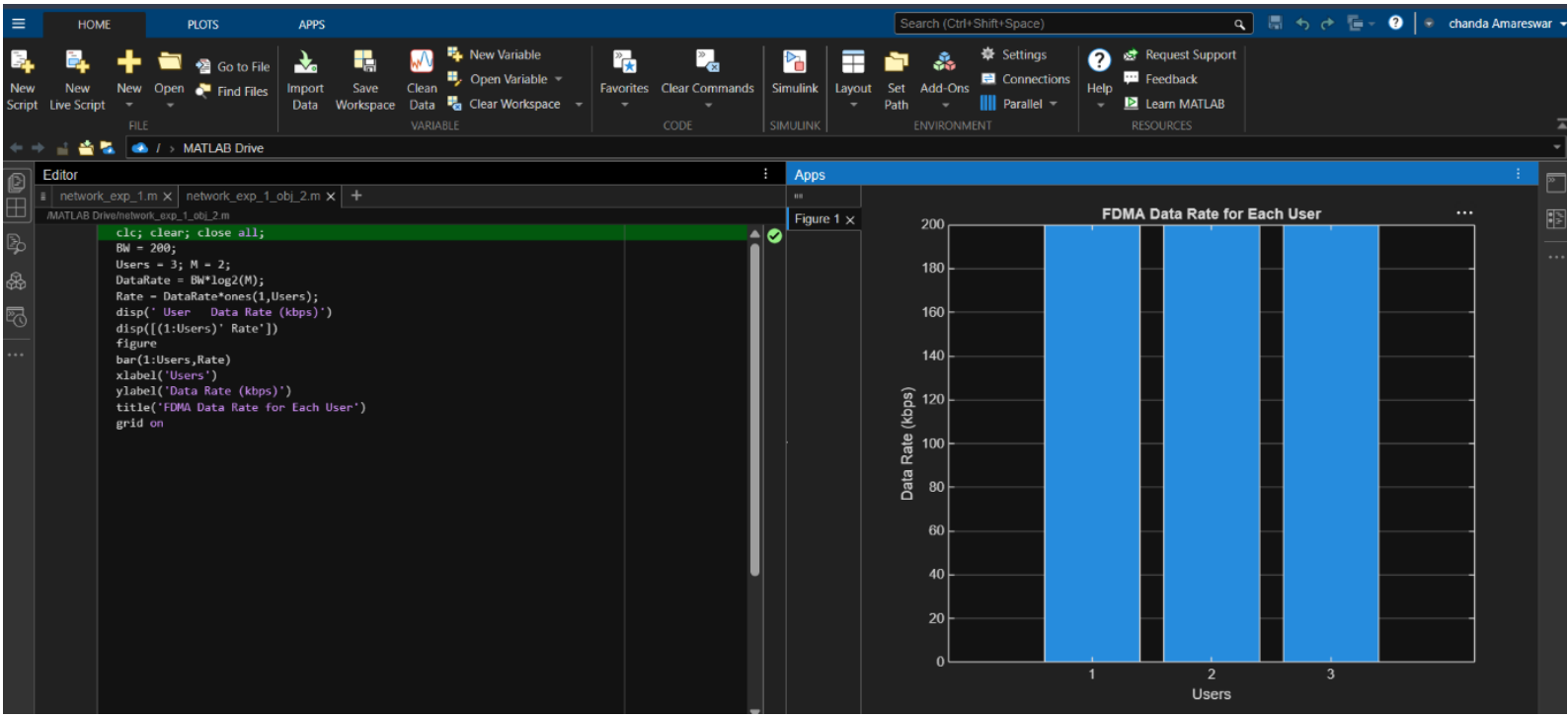


```
ylabel('Channel Bandwidth (kHz)'),
title('Channel Bandwidth per User'
```

EXPERIMENT 01(OBJ-2):-

PROGRAM CODE:-

```
clc; clear; close all;
 BW = 200;
 Users = 3; M = 2;                  % Channel Bandwidth (kHz) %
BPSK Modulation DataRate = BW*log2(M);       % Data Rate (kbps)
Rate = DataRate*ones(1,Users);
disp(' User   Data Rate (kbps)')
disp([(1:Users)' Rate'])
figure
bar(1:Users,Rate)
xlabel('Users')
 ylabel('Data Rate (kbps)')
 title('FDMA Data Rate for Each User')
grid on
```



EXPERIMENT 01(OBJ-3):-

PROGRAM CODE:-

```
clc; clear; close all;
 Ps = 1;
Pn = [0.1 0.05 0.01];
Users = 1:3;
SNR = 10*log10(Ps./Pn);
disp('User   Noise Power(W)   SNR(dB)')
disp([Users' Pn' SNR'])
figure
 bar(Users,SNR)
xlabel('Users')
ylabel('SNR (dB)')
```

title('Signal-to-Noise Ratio of FDMA Users')

grid on

